

Individual Contribution Assessment

Lebanese University Faculty of Engineering

Project Title: Smart Dual-Axis Solar-Tracking, Environmental Telemetry, and Automated Cleaning System

Course: Sensors

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1. Individual Contribution Summary

Although our engineering project was completed as a collaborative group effort, my personal responsibilities were highly distinct, bridging hands-on hardware optimization with total ownership of the system's software architecture. I served as the **software developer and system integrator** for the project, dedicating the majority of my efforts to writing, debugging, testing, and optimizing the comprehensive codebase. Additionally, I supported the team with crucial hardware tasks including precision soldering, physical installation of key sensors, and mechanical troubleshooting.

Primary Focus: Software Architecture and Engineering

As the team member focused on code generation and firmware diagnostics, I successfully engineered a fully integrated, multi-threaded logical flow. My key software achievements include:

- **LCD Memory Pointer Resolution:** I resolved a critical memory corruption error where our LiquidCrystal_I2C display outputted scrambled ("alien language") symbols. I diagnosed this as an overflow caused by attempting to print a four-row data structure on a physical 16x2 display, and engineered a consolidated, auto-refreshing two-row layout.
- **Software Debouncing Filters:** I mitigated erratic flowmeter calculations (artificial "ghost readings" up to 248 L/min) caused by contact bounce from our mechanical relay module and electromagnetic interference from the DC motor. I implemented a 2-millisecond software blanking window directly inside the interrupt service routine (ISR) to ignore physically impossible high-speed pulse spikes.

Secondary Focus: Hardware Tweaking, Assembly, & Calibration

In addition to software execution, I contributed directly to the physical assembly and calibration of our prototype:

- **Precision Soldering and Wiring:** I wired and soldered the essential analog paths for our four-photoresistor (LDR) voltage divider array to ensure noise-free signal delivery to the microcontroller.
- **Mechanical Troubleshooting and Realignment:** I diagnosed and fixed a severe mechanical stall on our vertical tilt servo by manually adjusting the physical gear alignment, resolving a mechanical binding issue that was causing the motor to draw excess current and overheat.
- **Sensor Integration and Potentiometer Calibration:** I mounted the environmental sensors (DHT22, HX711 Load Cell, IR Obstacle/Dust sensor, and flowmeter) to the physical chassis and calibrated the IR sensor's onboard hardware potentiometer to trigger specifically when actual dust accumulated.

Task Breakdown & Ownership

Project Domain	Specific Tasks Performed	Personal Ownership %
Software Engineering	Writing tracking algorithms, asynchronous loops, safety interlocks, debugging I2C memory, and implementing ISR debouncing filters.	90%
Hardware Assembly	Soldering LDR voltage dividers, wire routing, and mounting chassis sensors.	30%
Calibration & Diagnostics	Fine-tuning IR sensor potentiometers, manual servo realignment, and verifying voltage divider node behaviors.	50%

2. Evidence of Work Performed

My complete ownership of the system's development, code design, and real-time debugging is substantiated by several tangible parameters:

- Integrated Development & Diagnostic Workstation:** In the submitted project demonstration and testing videos, the personal laptop displaying the running integrated system code, executing compilation steps, and showing real-time diagnostic printouts via the Arduino Serial Monitor debugger is mine. This machine served as the central integration and verification station for our physical system.
- Iterative Diagnostic Codebases:** Before implementing the final integrated software, I wrote and independently validated several modular diagnostic sketches. These individual sketches—which I used to verify isolated component hardware—include raw LDR analog value readers, DHT22 serial drivers, independent flowmeter pulse counters, and servo-sweep test sketches.

3. Personal Reflection

Participating in this engineering sensors project has been a transformative experience, offering an invaluable bridge between theoretical calculations and the unpredictable dynamics of physical systems. Applying **fundamental circuit analysis theories** to the four-channel LDR voltage divider array showed how node-voltage equations translate into analog signals. Connecting abstract math with physical measurements made my engineering curriculum feel far more concrete.

Analyzing the **exact mechanisms behind our hardware components**—from DC motor inductive brushes to flowmeter Hall-effect pinwheels and servo mechanics—deepened my perspective. Seeing a stalled servo generate excessive thermal energy or a relay click cause an inductive power rail spike proved software does not exist in a vacuum. These challenges forced a holistic, systems-level engineering approach. Resolving real-world issues through custom software filters, power isolation, and mechanical calibrations significantly improved my confidence in physical computing, embedded systems, and collaborative design.